

Suggested Mark Scheme

- 1 (a) Complete Table 1.1 to show two differences between the three classes of macromolecules.

Table 1.1

Macromolecules	Polysaccharides	Proteins	Nucleic Acids
Monomers	<i>monosaccharides / glucose</i>	<i>amino acids</i>	<i>nucleotides</i>
Names of bonds linking monomers	<i>glycosidic</i>	<i>peptide</i>	<i>phosphodiester</i>

[2]

- (b)(i) State two structural differences between cellulose and starch.

glucose monomer: cellulose → β -glucose, starch → α -glucose;

bonds: starch → α -1,4 glycosidic bonds & α -1,6 glycosidic bonds (branching involved) , cellulose → only β -1,4 glycosidic bonds (no branching);

monomers orientation: cellulose → alternate glucose monomer inverted 180° , starch → adjacent glucose monomer in the same orientation;

location of hydroxyl group: cellulose → faces inwards forming H-bds with parallel chains, starch → faces outwards so that able to be stored in aqueous env of stroma;

[2]

- (ii) Explain how these differences allow them to serve different functions in a plant cell.

starch: helical struc/ highly branched → compact & good for storage, easily hydrolysed to release glucose, thus energy;

starch: insoluble in water, good as storage material without exerting osmotic pressure;

cellulose: H-bonds btw neighbouring chains (cross linking) → structural purpose, high tensile strength of cell wall;

[2]

- (c) Although cellulose and collagen belong to different classes of biomolecules, yet they serve similar function. Describe one important similarity that these two molecules share that allow them to serve their function well.

linear, straight chains;

ability of the chains to associate with other chains (forming of bonds between parallel chains;;

both are insoluble in water;

structural support; [2]

- (d) Suggest why there seems to be a greater diversity in terms of the types of proteins compared to polysaccharides.

greater variety/ at least 20 different aas/ variety due to side chain or R grps;

bonds – H, ionic, disulphide, hydrophobic interxns;

structure – 1°, 2°, 3° & 4°; [2]

[Total: 10]

- 2 The amino acid sequences of the two proteins (enzyme **X** and **Y**) were determined. Each contained 138 amino acids joined together in a single chain. The only differences in sequence were noticed in a central part of the proteins. In Table 2.1, the amino acids concerned are given, together with their numbered individual positions in the proteins. The sequence of 'bases' (nucleotides) corresponding to this central sequence is given for protein **X**.

Table 2.1

Protein X	86	87	88	89	90	91	92	93	
N terminal	Lys	Ser	Pro	Ser	Leu	Asn	Ala	Ala	C terminal
DNA sequence	TTT	TCA	GGT	AGT	GAA	TTA	CGA	CGA	
mRNA sequence									

- (a) Enter the appropriate mRNA sequence for protein **X** in Table 2.1. [1]

AAA, AGU, CCA, UCA, CUU, AAU, GCU, GCU

- (b) Describe briefly how the DNA sequence is converted to mRNA.

TF binds to TATA box;

RNA pol recognizes complex & binds to promoter region;

DNA unwound & coding/sense strand used as template;

add compl ribonucleotides via compl base pairing;

A=U, C≡G, T=A using H-bds;

transcription on 5' to 3' direction;

s/s mRNA formed;

[3]

Further investigations showed that the enzyme **Y** was a result of **two** mutations in the DNA coding for enzyme **X**. In the first mutation, a single nucleotide was **deleted** (lost) from the DNA. In the second mutation, a single nucleotide was **inserted** (added) to the DNA.

Table 2.2

Amino Acid	mRNA codons
Lys	AAA
Ser	AGU or UCA
Pro	CCA
Leu	CUU or UUA
Asn	AAU
Ala	GCU
Val	GUC
His	CAU or CAC
Met	AUG

- (c) Using the genetic codes given in Table 2.2, state
- (i) the sequence of mRNA codons for protein Y in Table 2.3;

Table 2.3

Protein Y	86	87	88	89	90	91	92	93	
N terminal	Lys	Val	His	His	Leu	Met	Ala	Ala	C terminal
mRNA sequence									

AAA, GUC, CAU, CAC, UUA, AUG, GCU, GCU; [1]

- (ii) the **DNA base** that was deleted in the first mutation;
T; [1]
-

- (iii) the **DNA base** that was inserted in the second mutation.
C; [1]
-

- (d) In another mutation, a single nucleotide was substituted in the DNA sequence that code for protein Y. However, there was no change in the activity of the enzyme.

Explain, with the aid of relevant examples from Table 2.2, why this is the case.

due to silent mutation;

due to degeneracy of genetic code, wobble base phenomenon;

e.g. in histidine, which is coded by both GTA & GTG (DNA triplet), a single substitution of A to G still codes for histidine;

change of a single nucleotide pair may switch an aa but have little effect on prot;

no change in aa seq, no change in R grp of aa;

no change in bonding, thus folding & 3D struc. Preserved; [3]

[Total: 10]

- 3 In anaerobic respiration in muscle tissue the conversion of glucose to lactic acid can be represented as a four stage as in Fig. 3.1:

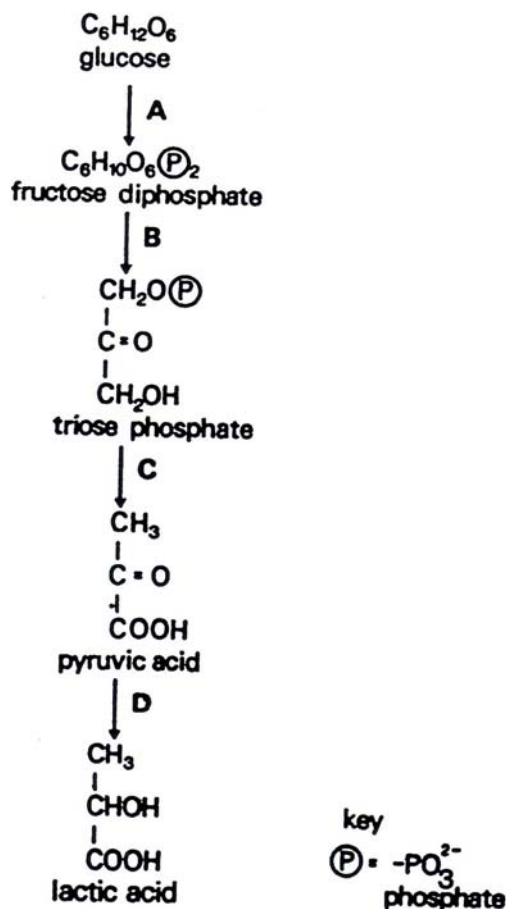


Fig. 3.1

- (a) Name the four stages in terms of the chemical processes occurring.

A phosphorylation of glucose

B cleavage / splitting

C dehydrogenation (oxidation) & dephosphorylation (substrate level phosphorylation)

D reduction / hydrogenation

[2]

- (b) Is there a net change in the concentrations of NAD and NADH₂ in the whole process? Explain your answer.

no;

during C, triose phosphate is oxidised / dehydrogenated;

NAD acts as H acceptor, forming NADH₂;

however, during stage D, pyruvate is reduced by accepting H from NADH₂ thus forming lactate;

thus regenerating NAD; [3]

- (c) If one molecule of glucose is converted to two molecules of lactic acid, what is the net synthesis of ATP? Explain your answer.

2 molecules of ATP;

in glycolysis, 4 mols of ATP are synthesized via substrate level phosphorylation from each glucose mol converted to pyruvate;

this is an offset agst 2 mols of ATP used for activation of glucose mol; [3]

- (d) In what ways is the ethanol formation different from that of lactate?

ethanol formation → yeast/plant, lactate formation → animal cells;

ethanol formation → CO₂ released (decarboxylation), lactate formation → no CO₂ released;

ethanol formation → involves 2 steps (formation of ethanal & formation of ethanol) with 2 enz (pyruvate decarboxylase & alcohol dehydrogenase, lactate formation → involves 1 step & 1 enzyme; [2]

[Total: 10]

- 4 A length of DNA from one of a pair of homologous chromosomes is shown in Fig. 4.1. The target sites of *EcoR*I are shown by arrows and the length of DNA between the target sites is given in kilobases (kb).

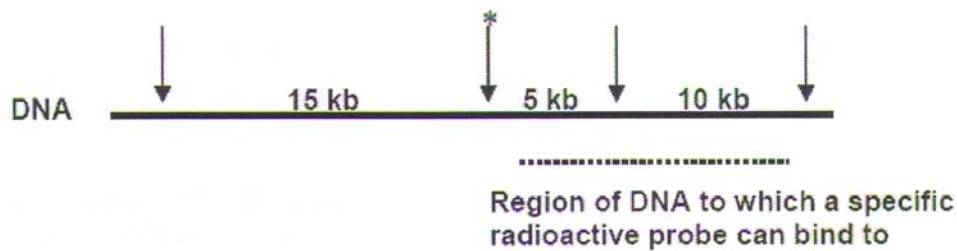


Fig. 4.1

A mutation alters one base of coding sequence at the site marked with an asterisk (*). DNA from two individuals, one who is normal and the other heterozygous for this mutation is cut with *EcoR*I and the DNA fragments separated by gel electrophoresis.

- (a) On Fig 4.2 below,
- Draw all the fragments of this length of DNA produced from the pair of homologous chromosomes in each of the individual tested and label the size of the fragments produced. [3]
 - Circle the fragments which will bind to the radioactive probe and appear as black bands on an autoradiography. [1]
 - Add an arrow to show the direction in which the fragments moved during electrophoresis. [1]

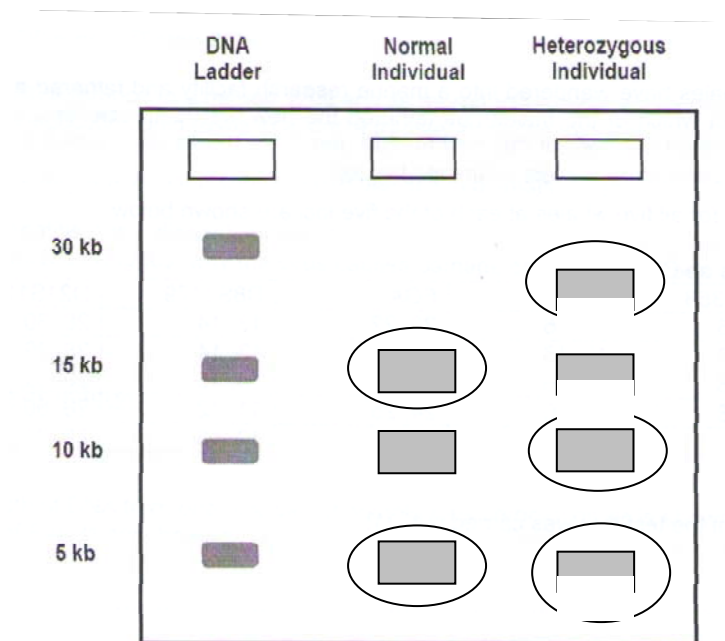


Fig. 4.2

- (b) State the purpose of DNA ladder.

each band of known size;

serves as a basis for comparison/ ref to size of DNA frag/ no. of repeats present in indiv present; [1]

- (c) Explain how the DNA fragments can be separated by electrophoresis.

a loading dye is added to DNA samples, which allows process of electrophoresis to be tracked/ helps DNA to sink into wells;

DNA samples loaded into wells near –ve electrode end of agarose gel;

gel is submerged in a buffer soln that will conduct electricity;

an electric current is applied through electrodes at opposite ends of gel;

-vely charged DNA frag migrate to +vely charged electrode/anode;

gel electrophoresis separates DNA frag by size/ mol wt;

with smallest frag moving fastest & furthest;

as smaller frag have lower resistance in moving thru pores of gel;

larger frag move slower due to higher resistance;

*invisible DNA bands can be viewed under UV light by staining gel [4]
with a methylene blue or ethidium bromide (EtBr);*

[Total: 10]

Section B**Answer EITHER 5 or 6 on the Answer Paper provided.**

- 5 (a)** Explain how homology (anatomical, embryological and molecular) supports Darwin's theory of natural selection. [10]

Homology

1. *presence of 2 or more species of a similar struc / characteristic;*
2. *derived from a recent common ancestor showing divergent evo;*

Anatomical homology

3. *similar physical struc such as bones, organs etc derived from a common ancestor when comparing btw orgs;*
4. *e.g. forelimbs of all mammals, human hand & flipper of whale;*
5. *same arrangement of bones from shoulder to tips of digits although appendages have diff fns: lifting, swimming;*
6. *pentadactyl limb is a homologous feature of all tetrapods, with diff forms in diff species;*
7. *descended from a common ancestor which had a basic form of pentadactyl limb → suggest common ancestr;*

Embryological homology

8. *comparison of early stages of animal devt not visible in adult org;*
9. *more closely related orgs are, longer embryological devt remains similar;*
10. *e.g. embryos of diff types of vertebrates → have a tail posterior to anus/pharyngeal (throat) pouches;*
11. *dev into homologous struc with diff fns, gill slits in fishes & parts of ears & throat in humans;*

Molecular homology

12. *orgs have similar genes & aa seq derived from a common ancestor;*
13. *DNA diffces are accumulated as descendants evolve / closely related species share a greater portion of their DNA;*
14. *e.g. analysis of cyt c, more diffces in nucleotides of DNA seq btw 2 species → less closely related they are;*

15. *due to a result of selection, changes in genes & gene freq, org dev differently;*
 16. *homologies show “descent with modification”*
 17. *comparisons of homologies btw species show how an ancestral homology in a popn may have been modified in descendent species through natural selection;*
 18. *homologies provide basis of comparison;*
 19. *homologous traits are compared btw popns or species as they are derived from a common ancestor → shows modification process from a basic ancestral form;*
 20. *unlike non-homologous (analogous) struc, e.g. wing of a bird & wing of insects, do not provide such a basis as they are derived from diff ancestors, although they may have a similar fn;*
- (b) Explain the unique features and normal functions of stem cells in a living organism, using appropriate examples to illustrate. [10]

Features:

1. *unspecialised & undifferentiated;*
2. *does not have any tissue specific struc that allows it to perform specialized fns;*
3. *capable of dividing & renewing themselves indefinitely;*
4. *stem cells can replicate many times via mitosis;*
5. *capable of differentiating into specialised cells under appropriate conditions;*
6. *induced by internal or external signals;*

Normal functions with e.g.:

7. *totipotent stem cells → from zygote or 4-day embryo (1st subsequent cleavage of zygote;*
8. *retains complete set of genetic info;*
9. *pot to dev into any cell type in adult body / entire org;*
10. *pluripotent stem cells → from inner cell mass of blastocysts;*
11. *pot to dev into any cell type except cells of extra-embryonic memb cells;*

12. e.g. embryonic stem cells → give rise to diff cell types/tissues/organs in developing foetus;
13. multipotent stem cells → found in tissues or organs;
14. pot to differentiate into a limited no. of cell types;
15. e.g. adult stem cells → hematopoietic stem cells in bone marrow;
16. give rise to all cells of blood but not other types of cells e.g. erythrocytes, B lymphocytes, T lymphocytes, neutrophils, basophils, platelets;
17. non-dividing until activated by tissue injury;
18. for maintenance & repair of tissue that are lost in disease, tissue injury & normal wear & tear in which they are found;
19. neuronal stem cells develops into neurons / nerve cells / skin stem cells develops into skin nerve cells / bone marrow stem cells develops into blood cells, liver cells, muscle cells;
20. unipotent (adult) stem cells capable of differentiating along 1 cell line same specific type e.g. liver cells;

[Total: 20]

6 (a) Compare the structure and organization of prokaryotic and eukaryotic genomes.

[10]

	<i>Prokaryote</i>	<i>Eukaryote</i>
<i>structure</i>	1. single, circular chromosome (d/s DNA)	1. multiple linear chromosomes (d/s DNA)
	2. located in nucleoid region	2. located in nucleus (bound by double memb of nuclear env)
	3. no telomeres, circular chromosome, no end replication problem	3. presence of telomeres @ ends of chromosomes, telomere shortening, with end replication problem
	4. multiple extrachromosomal plasmids	4. cpDNA &/or mtDNA
	5. smaller in size (10^5 to 10^7 base pairs)	5. larger in size (10^7 to 10^{11} base pairs)
	6. DNA (-vely charged), associated with histone like proteins (+vely charged) forming loops of DNA & supercoiling	6. DNA (-vely charged) wound around octamer of histone proteins (+vely charged) forming nucleosome, → folding

		<i>into chromatin fiber (solenoid)</i>
<i>organisation</i>	7. <i>genes (coding regions) organised into operons</i>	7. <i>genes not organised into operons, on diff chromosomes</i>
	8. <i>several structural genes under influence of single promoter</i>	8. <i>1 structural gene controlled by 1 promoter</i>
	9. <i>polycistronic → many prot encoded by 1 mRNA, many genes controlled by a single promoter</i>	9. <i>monocistronic → 1 mRNA encode only 1 prot, each gene has its own promoter and terminator</i>
	10. <i>absence of introns within genes</i>	10. <i>presence of introns within genes</i>
	11. <i>little non-coding regions, mostly coding regions, few repetitive seq</i>	11. <i>more non-coding than coding regions, many repetitive seq (e.g. centromere, telomere)</i>
	12. <i>AVP</i>	12. <i>AVP</i>

- (b) Describe how eukaryotes regulate gene expression at chromosomal and post transcriptional level. [10]

Feature	Eukaryotic Gene Expression
Chromosome modification	1. <i>via DNA methylation, histone acetylation & phosphorylation;</i> <u>DNA methylation</u> 2. <i>adding methyl grps to DNA converts C in CpG islands to 5-methyl C;</i> 3. <i>prevents binding of basal TFs to promoter → ↓ transcription;</i> <u>Histone acetylation</u> 4. <i>adding acetyl grps to histones , ↓ affinity of histone (+vely charged) for -vely charged DNA;</i> 5. <i>more relaxed chr conformation allowing access of TFs to promoter site → ↑ transcription;</i> <u>Phosphorylation</u> 6. <i>phosphorylation of lys & ser residues in histones → addition of -vely charged PO₄ grps neutralizes histones' +ve charge;</i> 7. <i>reduce their affinity for DNA → less coiling & easy access by RNA</i>

	<i>pol to promoters → ↑ transcription;</i>
Post transcriptional	8. <i>via 5' capping, 3' polyadenylation, RNA splicing, mRNA transport & mRNA editing; (any 3 correct)</i> <u>5' capping</u> 9. <i>adding of methylated guanine at 5' end of 1^o mRNA forming a 5' cap;</i> 10. <i>ensures mRNA's stability while it undergoes translation in prot synthesis in nucleus;</i> <u>3' polyadenylation</u> 11. <i>100 to 200 adenine (poly A) residues added to 3' end of 1^o mRNA;</i> 12. <i>poly(A) addition site consists of consensus seq, AAUAAA → defines position where poly(A) addition occurs;</i> 13. <i>facilitate export of mature mRNA from nucleus into cytosol;</i> 14. <i>protect mRNA from degradation by hydrolytic enzymes;</i> <u>RNA splicing</u> 15. <i>spliceosomes recognize highly conserved seq at 3' & 5' ends of introns (splice sites);</i> 16. <i>introns excised & exons spliced together forming mature mRNA;</i> 17. <i>alternative splicing → include diff exons together</i> 18. <i>↑ sing coding capacity of genome → ↑ diversity of prot formed;</i> <u>mRNA transport</u> 19. <i>formation of nuclear-export signal on mature mRNA with proteins</i> 20. <i>↑ mRNA tpt thru nuclear pores → ↑ chances of translation;</i> <u>mRNA editing</u> 21. <i>change in mRNA base to give rise to non-functional / diff prot i.e. regulate pdtn of diff prot;</i>